Logic and Set Theory - Representative Course Material

REMARK · math:prop:rk:course-map

Propositional Logic Course Map

Rigorous undergraduate core; classical, bivalent, truth-functional propositional logic.

Scope and source basis

This is a rigorous undergraduate treatment of classical, bivalent, truth-functional propositional logic. Its scope was checked against the inspected open textbooks *forall x: Calgary — A Free and Open Introduction to Formal Logic* and the Open Logic Project's *Propositional Logic*. These links are embedded here so the provenance remains visible after import into Study. The organization, formulations, proofs, and solutions below are independently synthesized for Study and for the explicitly declared calculus; they are not source quotations.

This course develops one formal system from syntax through metatheory. The stock of propositional variables is countably infinite. Formulas may use $\top$, $\bot$, $\neg$, $\wedge$, $\vee$, $\to$, and $\leftrightarrow$. Semantics is classical and two-valued. The proof system is the Fitch-style natural-deduction calculus stated in @math:prop:deduction:df:natural-deduction-calculus, including reductio ad absurdum, so it is a classical rather than intuitionistic calculus.

Read the modules in order:

1. Syntax and induction: formulas as finite structured objects.
2. Semantics: valuations, consequence, equivalence, and countervaluations.
3. Normal forms and resolution: representation and satisfiability methods.
4. Natural deduction: formal proofs with explicit assumption discharge.
5. Metatheory: soundness, completeness, compactness, consistency, and decidability.

The syntax-semantics distinction, formal derivability, soundness, completeness, consistency, and compactness developed here are foundations for later work on first-order logic, incompleteness, axiomatic set theory, and forcing. This statement is curricular: Gödel incompleteness and forcing are not results of propositional logic and are not covered in this course.

The scope is comprehensive for an undergraduate propositional-logic unit or course, but it deliberately does not claim to cover modal, intuitionistic, many-valued, relevance, or paraconsistent logics.

THEOREM · math:prop:normal-forms:th:canonical-normal-forms

Canonical DNF and CNF Theorem

Prerequisites: @math:prop:semantics:th:locality, @math:prop:normal-forms:df:literal-clause-term, and @math:prop:normal-forms:df:dnf-cnf

Let $P = \mathrm{At}(A) = \{p_1, \dots, p_n\}$. Using any fixed order of the finitely many assignments $a : P \to \{0,1\}$, the formula A is logically equivalent both to

$$\bigvee_{a(A)=1} m_a$$

and to

$$\bigwedge_{a(A)=0} k_a.$$

The finite conjunction and disjunction use the convention in @math:prop:normal-forms:df:literal-clause-term. The first formula is the canonical DNF and the second is the canonical CNF.

> ### Proof
>
> Fix a total valuation v and let $b = v|_P$ be its finite restriction. For every assignment $a : P \to \{0,1\}$, the minterm property gives $v \vDash m_a$ exactly when $b = a$. Hence v satisfies the displayed DNF exactly when m_b occurs in it, which is exactly when $b(A) = 1$. By the finite-assignment evaluation definition, $b(A) = \text{\llbracket}A\text{\rrbracket}_v$, so the DNF and A have the same value under v.
>
> Dually, v falsifies k_a exactly when $b = a$. The displayed CNF is therefore false under v exactly when its own maxterm k_b occurs, which is exactly when $b(A) = 0$, hence when $\text{\llbracket}A\text{\rrbracket}_v = 0$. Thus the CNF and A also agree under v. Since v was an arbitrary total valuation, both normal forms are logically equivalent to A.
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> If no finite assignment makes A true, the DNF is the empty disjunction $\bot$. If no finite assignment makes A false, the CNF is the empty conjunction $\top$. The same argument includes these boundary cases, including $P = \varnothing$.

PROBLEM · math:prop:metatheory:pb:compactness-graph-coloring

Use Compactness to Color an Infinite Graph

Prerequisites: @math:prop:normal-forms:df:literal-clause-term and @math:prop:metatheory:th:semantic-compactness

Let $G = (V, E)$ be a countable graph and let $k \geq 1$ be finite. Prove: if every finite induced subgraph of G has a proper k-coloring, then G has a proper k-coloring. Encode the claim propositionally and apply compactness.

Solution

For each $v \in V$ and color $i \in \{1, \dots, k\}$, use an atom $p_{v,i}$ meaning that vertex v has color i. Countability of V and finiteness of the color set make these atoms countable, so reindex them into the fixed stock $p_0, p_1, \dots$.

For each vertex v, define a right-associated finite disjunction recursively by

$$D_{v,k} = p_{v,k}, \qquad D_{v,i} = (p_{v,i} \vee D_{v,i+1}) \quad (1 \leq i < k).$$

Thus $D_{v,1}$ is the precise formula abbreviated by $\bigvee_{i=1}^{k} p_{v,i}$; the assumption $k \geq 1$ avoids an empty disjunction. Let Γ contain:

- $D_{v,1}$ for each vertex v;
- $(\neg p_{v,i} \vee \neg p_{v,j})$ for each vertex v and distinct colors i, j;
- $(\neg p_{u,i} \vee \neg p_{v,i})$ for each edge $\{u, v\} \in E$ and each color i.

The first family assigns at least one color, the second at most one, and the third different colors to adjacent vertices. Any finite $\Gamma_0 \subseteq \Gamma$ mentions only finitely many vertices W. By hypothesis, the finite induced graph on W has a proper k-coloring. Assign the corresponding atoms and extend arbitrarily to all other atoms; this valuation satisfies Γ_0. Hence every finite subset of Γ is satisfiable. Compactness gives a valuation satisfying all of Γ, and the unique true color atom for each vertex defines a proper k-coloring of G.

REMARK · math:fol:rk:course-guide

First-Order Logic Course Guide

Scope and conventions for the entire First-Order Logic branch.

Scope and source basis

This course develops classical, single-sorted first-order logic with equality over nonempty domains. It separates three levels throughout: an object-language formula such as φ, a syntactic judgment $\Gamma \vdash \varphi$, and a semantic claim $\Gamma \vDash \varphi$. The symbols $\vdash$ and $\vDash$ are never interchangeable by definition; soundness and completeness are the theorems that relate them.

The organization and conventions were checked against the Open Logic Project's *forall x: Calgary*, *Sets, Logic, Computation*, and *Intermediate Logic* course text. The prose and proofs here are independently synthesized. The first source supports introductory syntax, interpretations, identity, and natural deduction; the latter two support the Henkin construction, completeness, compactness, Löwenheim–Skolem, and the computability boundary.

Course path

The sequence is: formal syntax; Tarskian semantics; classical natural deduction; soundness and completeness; compactness and Löwenheim–Skolem; then normal forms, Skolemization, and computability limits. Function and relation symbols have fixed finite arities, equality is logical identity, and variable assignments are used for open formulas. Bound variables may be renamed only capture-free. Skolemization preserves satisfiability in an expanded language, not logical equivalence.

A complete first course should combine statement recall with actual derivations, countermodels, and metatheoretic proofs. The worked problems here are representative rather than an exhaustive exercise bank.

THEOREM · math:fol:metatheory:th:rule-soundness

Local Soundness of Natural-Deduction Rules

Each rule preserves semantic consequence relative to the open assumptions of its premise subderivations.

For each rule instance, suppose every premise occurrence is semantically entailed by the assumptions open at that point in its premise subderivation. Then the conclusion is semantically entailed by the assumptions that remain open after the rule's designated discharges.

Proof

Conjunction introduction and elimination, implication elimination, and disjunction introduction follow immediately from the classical truth conditions, with the premise contexts combined when necessary. For implication introduction, suppose $\Delta \cup \{\varphi\} \vDash \psi$. If $\mathfrak{M}, g \vDash \Delta$ and $\mathfrak{M}, g \vDash \varphi$, the premise consequence gives $\mathfrak{M}, g \vDash \psi$; hence $\mathfrak{M}, g \vDash \varphi \rightarrow \psi$. Thus $\Delta \vDash \varphi \rightarrow \psi$. For disjunction elimination, if $\Delta \vDash \varphi \vee \psi$, $\Sigma \cup \{\varphi\} \vDash \chi$, and $\Pi \cup \{\psi\} \vDash \chi$, then any $\mathfrak{M}, g$ satisfying $\Delta \cup \Sigma \cup \Pi$ satisfies one of φ, ψ and the corresponding branch consequence yields χ. Falsity elimination is vacuous because no assignment satisfies $\bot$. For classical reductio, if $\Delta \cup \{\neg\varphi\} \vDash \bot$ and some $\mathfrak{M}, g \vDash \Delta$ made φ false, classical semantics would make $\neg\varphi$ true, contradicting the premise consequence; therefore $\Delta \vDash \varphi$.

For universal elimination, $\Delta \vDash \forall x\varphi$ implies that any $\mathfrak{M}, g \vDash \Delta$ satisfies φ at $g[x \mapsto t^{\mathfrak{M}}[g]]$; semantic substitution gives $\mathfrak{M}, g \vDash \varphi[t/x]$. Existential introduction is the converse use of semantic substitution with witness $t^{\mathfrak{M}}[g]$. For universal introduction, suppose $\Delta \vDash \varphi$ and x is free in no member of Δ. Fix $\mathfrak{M}, g \vDash \Delta$ and arbitrary $a \in M$. By coincidence, $\mathfrak{M}, g[x \mapsto a] \vDash \Delta$, so the premise consequence gives $\mathfrak{M}, g[x \mapsto a] \vDash \varphi$. Since a was arbitrary, $\mathfrak{M}, g \vDash \forall x\varphi$.

For existential elimination, suppose $\Delta \vDash \exists x\varphi$ and $\Sigma \cup \{\varphi[y/x]\} \vDash \psi$, where y satisfies the stated freshness conditions. Fix $\mathfrak{M}, g \vDash \Delta \cup \Sigma$ and choose $a \in M$ witnessing $\exists x\varphi$. Put $h = g[y \mapsto a]$. Freshness and coincidence preserve every formula in $\Delta \cup \Sigma$; because y does not occur in the existential premise, semantic substitution gives $\mathfrak{M}, h \vDash \varphi[y/x]$. The branch consequence yields $\mathfrak{M}, h \vDash \psi$, and because y is not free in ψ, coincidence gives $\mathfrak{M}, g \vDash \psi$. Thus $\Delta \cup \Sigma \vDash \psi$. Equality introduction uses reflexivity of identity. Equality elimination is sound because equal terms have the same denotation, and semantic substitution shows that replacing one by the other preserves the formula's truth.

THEOREM · `math:fol:metatheory:th:completeness`

Gödel Completeness Theorem for First-Order Logic

Strong completeness; first stated for a theory Γ and a sentence φ.

For every set Γ of sentences and sentence φ, if $\Gamma \vDash \varphi$, then $\Gamma \vdash \varphi$. Together with soundness, $\Gamma \vdash \varphi$ if and only if $\Gamma \vDash \varphi$. The same result holds for arbitrary formulas under assignment semantics.

> ### Proof
>
> For sentences, argue contrapositively. If $\Gamma \nvdash \varphi$, then $\Gamma \cup \{\neg\varphi\}$ is consistent. Otherwise implication introduction would give $\Gamma \vdash \neg\neg\varphi$, and classical double-negation elimination would give $\Gamma \vdash \varphi$. By model existence there is $\mathfrak{M} \vDash \Gamma \cup \{\neg\varphi\}$, so $\mathfrak{M}$ satisfies every premise and falsifies φ. Hence $\Gamma \nvDash \varphi$. This proves $\Gamma \vDash \varphi \Rightarrow \Gamma \vdash \varphi$; soundness gives the reverse implication.
>
> For arbitrary formulas, choose a distinct new constant c_x for every variable free in a member of $\Gamma \cup \{\varphi\}$, and replace each such free x uniformly by c_x. Call the resulting sentences Γ^c and φ^c. Assignments in an L-structure correspond exactly to expansions interpreting the constants c_x, so $\Gamma \vDash \varphi$ iff $\Gamma^c \vDash \varphi^c$. The sentence case gives a finite derivation $\Gamma^c \vdash \varphi^c$. Before translating it back, alpha-rename all bound variables and rule eigenvariables away from the finitely many original free variables used in that derivation. Uniformly replace each c_x by x. Every rule instance and side condition is then preserved, yielding $\Gamma \vdash \varphi$. Thus the formula form follows without identifying $\vdash$ with $\vDash$.

PROBLEM · math:fol:syntax:pb:symbolize-nested-quantification

Symbolize Nested Quantification

Domain: people and books. $S(x)$ means "x is a student"; $B(y)$ means "y is a book"; $R(x, y)$ means "x read y".

Symbolize: (a) every student read some book; (b) some book was read by every student; (c) exactly one student read every book. Explain the difference between (a) and (b).

Solution

(a) $\forall x(S(x) \rightarrow \exists y(B(y) \wedge R(x, y)))$.

(b) $\exists y(B(y) \wedge \forall x(S(x) \rightarrow R(x, y)))$.

For (c), let $\Phi(x)$ abbreviate $S(x) \wedge \forall y(B(y) \rightarrow R(x, y))$. Then $\exists x(\Phi(x) \wedge \forall z(\Phi(z) \rightarrow z = x))$. Expanding Φ gives an ordinary first-order sentence.

In (a), the chosen book may depend on the student. In (b), one book must work for every student. Therefore (b) implies (a): use the common book as each student's witness. The converse fails when two students read different books and no one book is read by both. It also fails in a nonempty domain containing a nonstudent but no students or books: (a) is vacuously true while (b) is false.

REMARK · math:set:rk:course-guide

Set Theory Course Guide

Upper-division undergraduate scope; classical first-order logic with equality is assumed.

Scope and source basis

This is a rigorous first course in axiomatic set theory. It begins with the one-sorted first-order language whose only nonlogical symbol is membership, develops the axioms of Zermelo--Fraenkel set theory, and adds the Axiom of Choice only when it is explicitly invoked.

The organization was independently synthesized after inspecting four complementary, authoritative course benchmarks: Tim Button and the Open Logic Project's Set Theory: An Open Introduction; Richard Zach and the Open Logic Project's Sets, Logic, Computation; Yiannis Moschovakis's UCLA M114S Introduction to Set Theory; and the University of Cambridge Foundations preparation guide. Patrick Lutz's detailed introductory set-theory lecture sequence and exams provided a second first-course sequencing check. The prose, theorem statements, proofs, and solutions here are newly written rather than copied from those sources.

The default level is upper-division undergraduate: more rigorous than the set chapter of a discrete-mathematics survey, but deliberately short of a graduate independence course. The prerequisites are proof fluency, ordinary mathematical induction, and first-order syntax and semantics. The course rebuilds induction internally once ω has been constructed.

Course plan

The order is deliberate: formal setting and axioms; constructions; relations and functions; natural numbers and ordinary recursion; cardinality and countability; orders and well-orders; ordinals and transfinite methods; Hartogs' theorem; equivalent forms of Choice; cardinal arithmetic; Foundation and the cumulative hierarchy; and finally a precise survey boundary around the Continuum Hypothesis and independence.

Unless an entry says otherwise, proofs are carried out in ZF. After the Axiom of Choice is introduced, statements requiring it are marked **(ZFC)**. A *class* is always metalinguistic shorthand for the sets satisfying a formula; this course does not silently move to a two-sorted class theory.

The course proves the core first-course results. It states, but does not pretend to prove, the Gödel--Cohen independence results. Constructibility, forcing, large cardinals, and advanced infinitary combinatorics belong to a graduate sequel. Long constructions of the integers, rationals, and reals are also omitted; only the set-theoretic facts needed for cardinality are used.

THEOREM · math:set:ordinals:th:transfinite-recursion

Transfinite Recursion

Let θ be an ordinal, fix set parameters $\vec{p}$, and let $\varphi(h, y, \vec{p})$ be a first-order formula. Suppose that whenever h is a function whose domain is an ordinal below θ, there is exactly one set y satisfying $\varphi(h, y, \vec{p})$. Then there is a unique function F with domain θ such that

$$\varphi(F \restriction \alpha, F(\alpha), \vec{p}) \quad \text{for every } \alpha < \theta.$$

The formula-definability and set-parameter requirements are what permit the uses of Replacement; an arbitrary external class assignment is not covered by this theorem as stated in ZF.

Proof

Call f an approximation through α if $\mathrm{dom}(f) = \alpha + 1$ and $\varphi(f \restriction \beta, f(\beta), \vec{p})$ holds for every $\beta \leq \alpha$.

Two approximations agree on their common domain. Otherwise their least point of disagreement γ would have identical restrictions below γ. Both values at γ satisfy φ for that same restriction, so the assumed uniqueness would force them to be equal.

We prove by transfinite induction that an approximation through every $\alpha < \theta$ exists. Assume approximations exist through all $\beta < \alpha$. They are unique by the preceding paragraph, and "f is the approximation through β" is first-order definable from φ and $\vec{p}$. Replacement therefore collects them into a set. Their union h is a function with domain α because they are compatible. Let y be the unique set with $\varphi(h, y, \vec{p})$, and append $\langle \alpha, y \rangle$. The resulting function is an approximation through α. This also handles 0 and limit stages: at 0 the prior union is empty, while at a limit it glues all earlier approximations.

Replacement now collects the unique approximation through each $\alpha < \theta$; their union is a function F on θ, and the construction gives the required instances of φ. If F' also satisfies them, a least point at which F and F' differ yields the same uniqueness contradiction used for approximations. Hence F is unique.

DEFINITION · math:set:cardinal-arithmetic:df:cardinal-operations

Cardinal Addition, Multiplication, and Exponentiation

(ZFC convention for the full package.) For cardinals κ and λ, define

$$\begin{aligned}\kappa + \lambda &= |(\{0\} \times \kappa) \cup (\{1\} \times \lambda)|,\\ \kappa \cdot \lambda &= |\kappa \times \lambda|,\\ \kappa^{\lambda} &= |\operatorname{Fun}(\lambda, \kappa)|,\end{aligned}$$

where $\operatorname{Fun}(\lambda, \kappa)$ is the set of functions from λ to κ. The tagged union in the first line keeps the summands disjoint.

For a set-indexed family of cardinals $\langle \kappa_i : i \in I \rangle$, define

$$\sum_{i \in I} \kappa_i = |\{\langle i, \alpha \rangle : i \in I \text{ and } \alpha < \kappa_i\}|$$

and

$$\prod_{i \in I} \kappa_i = |\{f : \operatorname{dom}(f) = I \text{ and } f(i) < \kappa_i \text{ for every } i \in I\}|\,.$$

Binary cardinal addition and multiplication of well-orderable sets are already well-defined in ZF: the displayed tagged union and Cartesian product can be explicitly well-ordered. In contrast, even when κ and λ are initial ordinals, ZF alone need not prove that $\operatorname{Fun}(\lambda, \kappa)$ is well-orderable, and an indexed sum or product over an arbitrary set I presents the same issue. The initial-ordinal values in the full definition above therefore use ZFC. These are cardinal operations: only bijection type matters. They must not be confused with ordinal operations, which retain order-type information and need not commute.

THEOREM · math:set:cardinal-arithmetic:th:konig-theorem

König's Theorem

(ZFC) If $\kappa_i < \lambda_i$ for every $i \in I$, then

$$\sum_{i \in I} \kappa_i < \prod_{i \in I} \lambda_i.$$

Proof

Treat each cardinal as its initial ordinal. There is an injection from the disjoint sum to the product: send (i, α) to the function that has value α at coordinate i and value κ_j at every coordinate $j \neq i$. This is legitimate because $\alpha < \kappa_i < \lambda_i$ and $\kappa_j < \lambda_j$. Images of different tagged summands differ at one of their tags, since the default value κ_i is outside the range κ_i.

There is no surjection in the other direction. Let

$$F : \bigsqcup_{i \in I} \kappa_i \longrightarrow \prod_{i \in I} \lambda_i$$

be any function. For each i, the set

$$C_i = \{F(i, \alpha)(i) : \alpha < \kappa_i\}$$

has cardinal at most κ_i, so it is a proper subset of λ_i. Because λ_i is an ordinal, define without making any further choice

$$b_i = \min(\lambda_i \setminus C_i).$$

Replacement collects these uniquely defined values into the function $b = (b_i)_{i \in I}$. It lies in the product and differs from every $F(i, \alpha)$ at coordinate i. Hence F is not surjective. (The theorem is stated in ZFC so that the indexed cardinal operations have initial-ordinal values; the diagonal selection itself uses least elements, not an additional appeal to Choice.)

The injection plus nonexistence of a reverse equinumerosity gives the strict inequality.

REMARK · math:set:continuum-independence:rk:advanced-boundary

Boundary of a First Set Theory Course

The following are important follow-on subjects, not prerequisites silently smuggled into this course:

- the constructible universe L, absoluteness, reflection, and inner models;
- forcing, generic extensions, Boolean-valued models, and independence proofs;
- regular and singular cardinals, stationary and club sets, and infinitary combinatorics;
- large-cardinal axioms and consistency strength;
- descriptive set theory and determinacy;
- fine structure of fragments of Choice.

A graduate set-theory sequence usually begins by reviewing ZFC, ordinals, cardinals, transfinite recursion, Hartogs' theorem, Choice, and rank—the material developed here—then adds model-theoretic and metamathematical machinery. Naming an advanced result is not the same as proving it, and a relative-consistency result must never be reported as an unconditional theorem about the intended universe of sets.